

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}$$

$$\max f(x) = 4x_1 + 3x_2$$

$$x_1 + x_2 \leq 9$$

$$2x_1 + x_2 \leq 12$$

$$x_1 + 3x_2 \leq 18$$

$$x_1, x_2 \geq 0$$

Metodo del simplesso:

	x_1	x_2	s_1	s_2	s_3	b
s_1	1	1	1	0	0	9
s_2	2	1	0	1	0	12
s_3	1	3	0	0	1	18
	-4	-3	0	0	0	0

Entra x_1 ed esce s_2 :

	x_1	x_2	s_1	s_2	s_3	b
s_1	1	1	1	0	0	9
s_2	2	1	0	1	0	12
s_3	1	3	0	0	1	18
	-4	-3	0	0	0	0

$$R2 \leftarrow R2 \cdot \frac{1}{2}$$

	x_1	x_2	s_1	s_2	s_3	b
s_1	1	1	1	0	0	9
s_2	1	$\frac{1}{2}$	0	$\frac{1}{2}$	0	6
s_3	1	3	0	0	1	18
	-4	-3	0	0	0	0

$$R1 \leftarrow R1 - R2$$

$$\begin{array}{cccc|c} 1 & 1 & 1 & 0 & 0 & 9 \\ 1 & \frac{1}{2} & 0 & \frac{1}{2} & 0 & 6 \\ \hline 0 & \frac{1}{2} & 1 & -\frac{1}{2} & 0 & 3 \end{array}$$

↓

	x_1	x_2	s_1	s_2	s_3	b
s_1	0	$\frac{1}{2}$	1	$-\frac{1}{2}$	0	3
s_2	1	$\frac{1}{2}$	0	$\frac{1}{2}$	0	6
s_3	1	3	0	0	1	18
	-4	-3	0	0	0	0

$$R3 \leftarrow R3 - R2$$

$$\begin{array}{cccc|c} 1 & 3 & 0 & 0 & 1 & 18 \\ 1 & \frac{1}{2} & 0 & \frac{1}{2} & 0 & 6 \\ \hline 0 & \frac{3}{2} & 0 & -\frac{1}{2} & 1 & 12 \end{array}$$

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	x_1	x_2	s_1	s_2	s_3	b
s_1	0	$\frac{1}{2}$	1	$-\frac{1}{2}$	0	3
s_2	1	$\frac{1}{2}$	0	$\frac{1}{2}$	0	6
s_3	0	$\frac{3}{2}$	0	$-\frac{1}{2}$	1	12
	-4	-3	0	0	0	0

Forma standard

$$\min f(x) = -4x_1 - 3x_2$$

$$x_1 + x_2 + s_1 = 9$$

$$2x_1 + x_2 + s_2 = 12$$

$$x_1 + 3x_2 + s_3 = 18$$

$$x_1, x_2, s_1, s_2, s_3 \geq 0$$

$$R4 \leftarrow R4 + R2 \cdot 4$$

$$\begin{array}{ccccc|c} -4 & -3 & 0 & 0 & 0 & 0 \\ 4 & 2 & 0 & 2 & 0 & 24 \\ \hline 0 & -1 & 0 & 2 & 0 & 24 \end{array}$$

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	x_1	x_2	s_1	s_2	s_3	b
s_1	0	$\frac{1}{2}$	1	$-\frac{1}{2}$	0	3
x_1	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	0	6
s_3	0	$\frac{5}{2}$	0	$-\frac{1}{2}$	1	12
	0	-1	0	2	0	24

Entra x_2 ed esce s_3 :

	x_1	x_2	s_1	s_2	s_3	b
s_1	0	$\frac{1}{2}$	1	$-\frac{1}{2}$	0	3
x_1	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	0	6
s_3	0	2	0	$-\frac{1}{2}$	1	12
	0	-1	0	2	0	24

$$R3 \leftarrow R3 \cdot \frac{2}{5}$$

	x_1	x_2	s_1	s_2	s_3	b
s_1	0	$\frac{1}{2}$	1	$-\frac{1}{2}$	0	3
x_1	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	0	6
s_3	0	1	0	$-\frac{1}{5}$	$\frac{2}{5}$	$\frac{24}{5}$
	0	-1	0	2	0	24

$$R1 \leftarrow R1 - R3 \cdot \frac{1}{2}$$

$$\begin{array}{ccccc|c} 0 & \frac{1}{2} & 1 & -\frac{1}{2} & 0 & 3 \\ 0 & \frac{1}{2} & 0 & -\frac{1}{10} & \frac{1}{5} & \frac{12}{5} \\ \hline 0 & 0 & 1 & \frac{2}{5} & -\frac{1}{5} & \frac{3}{5} \end{array}$$

↓

	x_1	x_2	s_1	s_2	s_3	b
s_1	0	0	1	$\frac{2}{5}$	$-\frac{1}{5}$	$\frac{3}{5}$
x_1	1	$\frac{1}{2}$	0	$\frac{1}{5}$	0	6
s_3	0	1	0	$-\frac{1}{5}$	$\frac{2}{5}$	$\frac{24}{5}$
	0	-1	0	2	0	24

$$R2 \leftarrow R2 - R3 \cdot \frac{1}{2}$$

$$\begin{array}{ccccc|c} 1 & \frac{1}{2} & 0 & -\frac{1}{2} & 0 & 6 \\ 0 & \frac{1}{2} & 0 & -\frac{1}{10} & \frac{1}{5} & \frac{12}{5} \\ \hline 1 & 0 & 0 & \frac{3}{5} & -\frac{1}{5} & \frac{18}{5} \end{array}$$

↓

	x_1	x_2	s_1	s_2	s_3	b
s_1	0	0	1	$\frac{2}{5}$	$-\frac{1}{5}$	$\frac{3}{5}$
x_1	1	0	0	$\frac{3}{5}$	$-\frac{1}{5}$	$\frac{18}{5}$
s_3	0	1	0	$-\frac{1}{5}$	$\frac{2}{5}$	$\frac{24}{5}$
	0	-1	0	2	0	24

$$R4 \leftarrow R4 + R3$$

$$\begin{array}{ccccc|c} 0 & -1 & 0 & 2 & 0 & 24 \\ 0 & 1 & 0 & -\frac{1}{5} & \frac{2}{5} & \frac{24}{5} \\ \hline 0 & 0 & 0 & \frac{9}{5} & \frac{2}{5} & \frac{144}{5} \end{array}$$

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	x_1	x_2	s_1	s_2	s_3	b
s_1	0	0	1	$\frac{2}{5}$	$-\frac{1}{5}$	$\frac{3}{5}$
x_1	1	0	0	$-\frac{1}{5}$	$-\frac{1}{5}$	$\frac{18}{5}$
x_2	0	1	0	$-\frac{1}{5}$	$\frac{2}{5}$	$\frac{24}{5}$
	0	0	0	$\frac{9}{5}$	$\frac{2}{5}$	$\frac{144}{5}$

$$x_1 = \frac{18}{5}$$

$$x_2 = \frac{24}{5}$$

$$\min f(x) = -4x_1 - 3x_2 = -4\left(\frac{18}{5}\right) - 3\left(\frac{24}{5}\right) = -\frac{72}{5} - \frac{72}{5} = -\frac{144}{5} \rightarrow \max f(x) = 4x_1 + 3x_2 = \boxed{\frac{144}{5}}$$